

RAID in Storage Systems

1 Introduction

RAID (**R**edundant **A**rray of **I**ndependent/**I**nexpensive **D**isks) is a method of combining multiple physical hard drives into a single logical unit to achieve:

- Higher performance (parallelism)
- Fault tolerance (redundancy)
- Increased storage efficiency

The effectiveness of RAID depends on the chosen **RAID level**. Below, we explore common RAID levels with diagrams, pros/cons, and detailed properties.

2 RAID Levels Overview

Each RAID level is a trade-off between speed, redundancy, and efficiency.

2.1 RAID 0 – Striping

Concept

Data is striped across multiple disks without any redundancy.

Pros

- Very high throughput (parallel disk usage).
- 100% storage utilization.

Cons

- No fault tolerance.
- A single disk failure = total data loss.

Details:

- **Speed:** Very high.
- **Safety:** 0 disk failures tolerated.
- **Utilization:** 100%.
- **Applications:** Video editing, temporary files.
- **Write Operation:** A block is written directly to the appropriate disk.

- **Cost per write:** 1 write per block.
- **Reason:** No redundancy or parity calculation is needed, so minimal overhead.

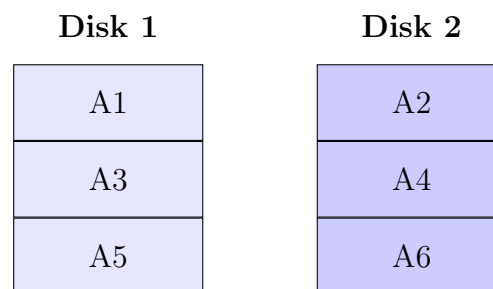


Figure 1: RAID 0 – Data Striping Across Two Disks

2.2 RAID 1 – Mirroring

Concept

Each disk stores an identical copy (mirror).

Pros

- Excellent redundancy.
- Fast read speed.

Cons

- Storage efficiency = 50%.
- Write operations are duplicated.

Details:

- **Speed:** Medium.
- **Safety:** 1 disk failure tolerated.
- **Utilization:** 50%.
- **Applications:** OS drives, databases.
- **Write Operation:** Every write updates both disks of the mirrored pair.
- **Cost per write:** 2 writes per block.
- **Reason:** To maintain redundancy, both disks must contain the same data. Reads can be from either disk, but writes are doubled.

Disk 1	Disk 2
A1	A1
A2	A2
A3	A3

Figure 2: RAID 1 – Mirroring (Data Duplicated Across Two Disks)

2.3 RAID 5 – Block-level Striping + Distributed Parity

Concept

Data and parity blocks are striped across all disks.

Pros

- No single parity bottleneck.
- Good balance of speed and safety.

Cons

- Write penalty due to parity updates.
- Rebuilds are slow after failure.

Details:

- **Speed:** Good (parallel reads).
- **Safety:** 1 disk failure tolerated.
- **Utilization:** $(n-1)/n$. *Explanation:* For n disks, 1 disk's worth of space is reserved for parity. Hence usable space = $n - 1$ disks.
- **Applications:** Web servers, general-purpose storage.
- **Write Operation (small write):**
 1. Read old data
 2. Read old parity
 3. Write new data
 4. Write new parity
- **Cost per write:** 4 I/O operations per small write.
- **Reason:** Parity must be updated for every write, so a single block update involves multiple disk operations. Large sequential writes are more efficient.

Disk 1	Disk 2	Disk 3
A1	A2	P1
P2	A4	A5
A3	P3	A6

Figure 3: RAID 5 – Distributed Parity Across Three Disks

2.4 RAID 6 – Dual Distributed Parity

Concept

Like RAID 5, but stores two parity blocks per stripe.

Details:

- **Speed:** Lower than RAID 5.
- **Safety:** 2 disk failures tolerated.
- **Utilization:** $(n - 2)/n$. *Explanation:* For n disks, 2 disks' worth of space is used for parity.
- **Applications:** Backup, archival storage.
- **Write Operation (small write):**
 1. Read old data
 2. Read old parity 1
 3. Read old parity 2
 4. Write new data
 5. Write new parity 1
 6. Write new parity 2
- **Cost per write:** Approximately 6 I/O operations per small write.
- **Reason:** Updating two parity blocks requires reading old parity values and writing updated parity after each data change.

Disk 1	Disk 2	Disk 3	Disk 4
A1	A2	P1	Q1
P2	Q2	A3	A5
Q3	P3	A4	A6

Figure 4: RAID 6 – Dual Distributed Parity Across Four Disks

2.5 RAID 10 – Nested RAID (1+0)

Concept

Combines RAID 1 (mirroring) with RAID 0 (striping).

Details:

- **Speed:** Very high.
- **Safety:** Multiple failures tolerated (if not in same pair).
- **Utilization:** 50%.
- **Applications:** Enterprise databases, mission-critical systems.
- **Write Operation:** Each block is written to both disks in a mirror (RAID 1), then striped across mirrors.
- **Cost per write:** 2 writes per block.
- **Reason:** Mirroring doubles the write operations, while striping does not add additional writes.

Disk 1	Disk 2	Disk 3	Disk 4
A1	A1	A2	A2
A3	A3	A4	A4

Figure 5: RAID 10 – Mirrored Striping Across Four Disks

3 Comparison of RAID Levels

RAID	Speed	Safety	Utilization	Cost/Write	Applications
RAID 0	Very High	None	100%	1 write per block	Scratch disks, temp data
RAID 1	Medium	1 disk	50%	2 writes per block (mirrored)	OS, DBs
RAID 5	Good balance	1 disk	$(n - 1)/n$	4 I/Os	General servers
RAID 6	Moderate	2 disks	$(n - 2)/n$	6 I/Os	Archival, backups
RAID 10	Very High	Multiple	50%	2 writes per block	Enterprise DBs

Table 1: Comparison of Modern RAID Levels

4 Legacy vs Modern RAID

Key Insight

- RAID 2, 3, 4 are obsolete (parity bottlenecks, ECC redundancy).
- RAID 5, 6, and 10 dominate modern servers.
- SSDs shift RAID strategies due to low latency.
- Cloud systems use erasure coding instead of RAID.